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Studio Session	5

Part A: URL of the Visualisation

https://ehoo0034.github.io/DV2_FIT2179_36448818/

Part B: Domain Description

Describe your domain chosen for DV II

Domain chosen is Healthcare Access in the KL-Selangor Transit Region. This project aims to investigate the spatial and operational efficiency of the rail transit network in supporting public access, especially the elderly to healthcare services. It highlights the disparities in the connectivity between dense urban areas and the needs of the elderly

Describe the Why and Who aspects of your DV II

Why

- To explore and evaluate if the development of transit rail networks in KL-Selangor successfully integrates social infrastructure
- To identify areas with high senior population density that simultaneously show high Isolation score, where senior population stay far awa from transit stations and do not have easy access to healthcare facilities
- To compare between different transit lines based on multi-dimensional performance metrics (hospital proximity, frequency of trains, accessibility of stations, interconnectivity, average fare), allowing users to understand which lines are most accessible

Who

- General Public: To visualize the accessibility of healthcare services and better route planning when visiting healthcare facilities
- Public Health authorities: To identify the demographic needs of seniors that reside far away from transit stations and healthcare facilities

- Urban Developers: To evaluate potential next phase of certain transit lines and how to integrate it into existing infrastructure

Describe the "What" aspect of your DV II

- Spatial data: GeoJSON district boundaries of the KL-Selangor region, point data for healthcare facilities, point data for transit station locations
    - JAKIM Malaysia District Boundaries GeoJSON (GitHub, 13 Oct 2022)
    - Humanitarian Data Exchange – Malaysia Health Facilities (OpenStreetMap, 8 Apr 2026 & 14 May 2026)
    - data.gov.my GTFS Static API – Prasarana Malaysia (accessed May 2026)

- Demographic data: Age and gender population data filtered to identify senior demographic as well as the overall population
    - Open DOSM – Population Table: Malaysia (OpenDOSM, 31 Jul 2025)
- Performance data: Aggregated metrics on ridership, transit line frequency, accessibility, fare prices, interconnectivity and travel mode
    - Ministry of Transport (MoT-data.gov.my, 12 May 2026)
    - RapidKL (myrapid.com.my, 17 May 2026)
    - Asian Transport Observatory (Dec 2024)

Describe the "How" aspect of your DV II

- Choropleth map visualizes senior population density across KL-Selangor districts
- Layered map with walking buffers identifies areas of transit-health connectivity
- Radar chart compares the transit line performance across five metrics
- Bar/Waffle chart quantify the healthcare facility mix and their proximity to transit stations
- Scatter plot correlates station ridership with healthcare facility density, showing how transit station traffic correlates to utility of medical s
- Heatmap compares senior population density against a calculated Isolation score, helping highlight districts that show a gap between transit and healthcare

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Part C: Sketch

file:///C:/Users/emily/OneDrive/Documents/FIT2179/DV2/DV2_Sketch.pdf

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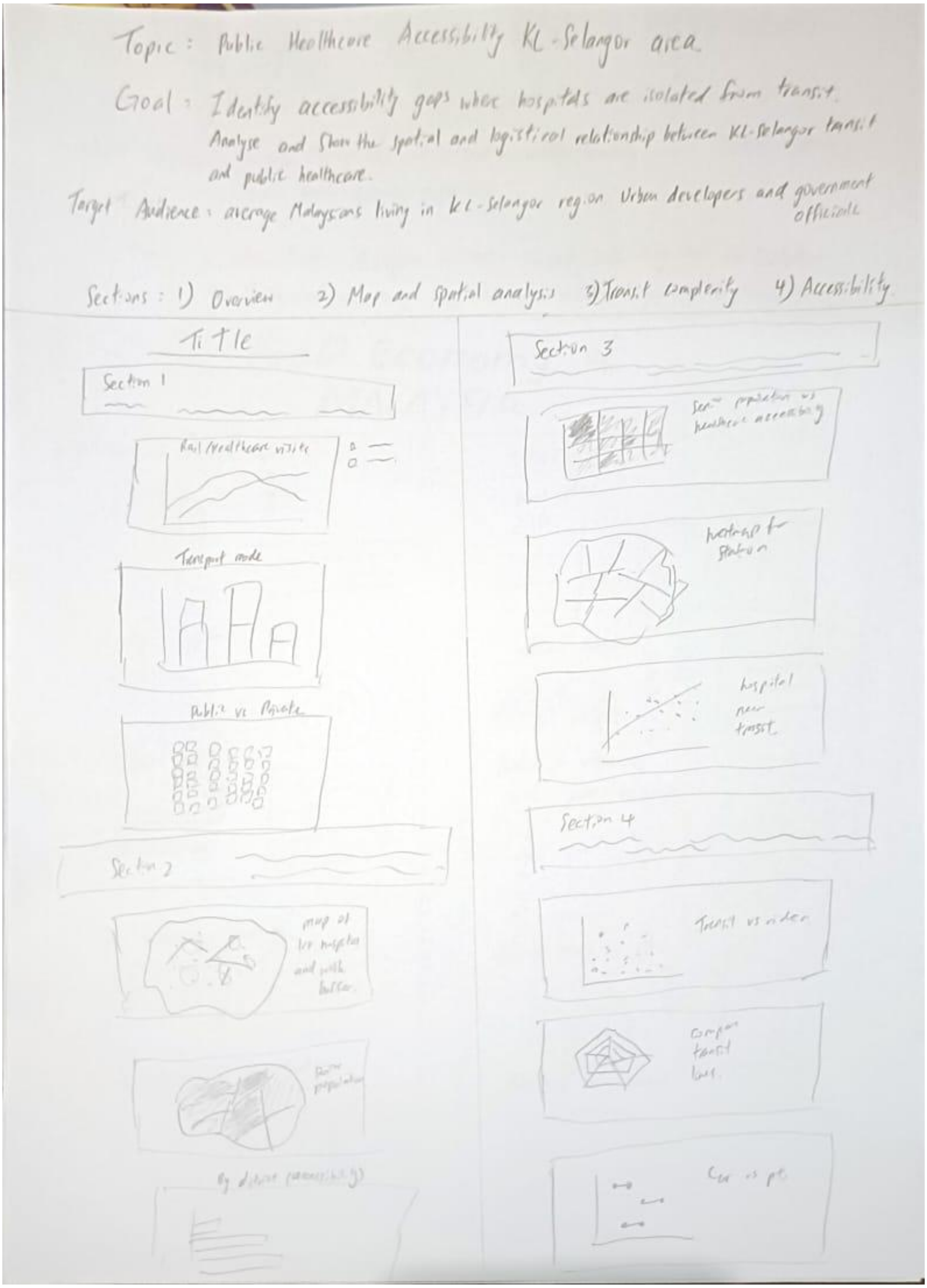
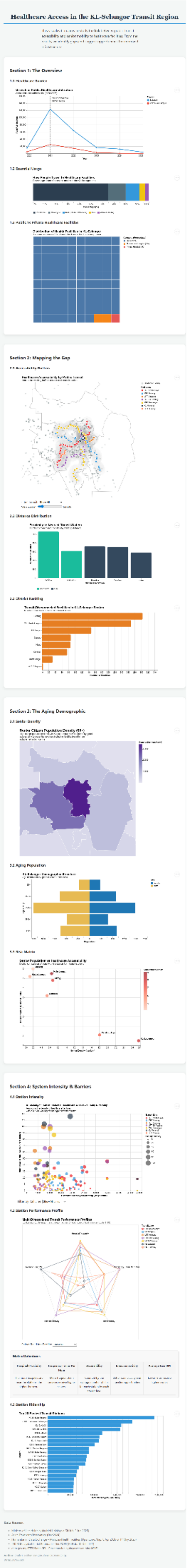

Topic: Public Healthcare Accessibility KL-Selangor area
Goal: Identify accessibility gaps where hospitals are isolated from transit.
Analyse and Show the spatial and logistical relationship between KL-Selangor transit and public healthcare.
Target Audience: average Malaysians living in KL-Selangor region Urban developers and government officials
Sections: 1) Overview 2) Map and spatial analysis 3) Transit complexity 4) Accessibility
Title
Section 1
Transport mode
Public vs Private
Section 2
Section 3
hospital near transit
Section 4

Part D: Full-size screenshot of your DV II


Healthcare Access in the KL-Selangor Transit Region

Gen-AI acknowledgement

I acknowledge the use of Google Gemini (https://gemini.google.com/app) to refine the academic language and accuracy of my own work. I used gemini to help refine my words for some explanations. I later paraphrased and made some edits to the output. I also used copilot and chatgpt to help with debugging for some of my charts. Additionally, I asked gemini how to upload files to github using terminal. This was because git had a 25mb drop file limit which one of my files exceeded.